Task Sheet Week 13 Day 2

Task I: Complete the comparison table below by analyzing abstract A and B.

Criteria	What to Look For?	Abstract A	Abstract B
Clear Problem Statement	Does the abstract clearly explain the research problem or issue being addressed?		
Well-Defined Purpose / Objective	Does it clearly state what the study aims to achieve?		
Methodological Detail	Does it mention dataset, tools, models, or techniques used in a specific and technical way?		
Use of Quantitative Results	Does it include clear numerical results (e.g., accuracy, %, performance scores)?		
Technical Specificity	Does it mention specific models, algorithms, datasets, or features instead of general terms?		
Information Clarity & Precision	Is the information clear, focused, and free from vague statements?		
Academic Tone & Formality	Is the writing formal, objective, and suitable for academic/research context?		
Research Insight / Conclusion Strength	Does it clearly explain what the results mean or contribute to the field?		

Abstract A
Financial fraud is a major problem that affects financial markets around the world. It is difficult to detect fraud because there are many more non-fraud companies than fraudulent ones. Many systems have been developed to help in detecting fraud, but most of them focus only on numerical financial data and do not consider textual information. In this study, a system is developed to improve fraud detection by using both numerical and textual data. The data was collected from Chinese companies, and different features were extracted from financial statements and management discussions. Deep learning models were applied and compared with other methods to see which works better. The results showed that deep learning performed better than traditional machine learning approaches. The study also found that using both numerical and textual data can improve detection. Overall, the proposed system can help in identifying financial fraud more effectively and can be useful for decision-making.

Abstract B
Financial fraud has extremely damaged the sustainable growth of financial markets as a serious problem worldwide. Nevertheless, it is fairly challenging to identify frauds with highly imbalanced dataset because ratio of non-fraud companies is very high compared to fraudulent ones. Intelligent financial statement fraud detection systems have therefore been developed to support decision-making for the stakeholders. However, most of current approaches only considered the quantitative part of the financial statement ratios while there has been less usage of the textual information for classifying, especially those related comments in Chinese. As such, this paper aims to develop an enhanced system for detecting financial fraud using a state-of-the-art deep learning models based on combination of numerical features that derived from financial statement and textual data in managerial comments of 5130 Chinese listed companies' annual reports. First, we construct financial index system including both financial and non-financial indices that previous researches usually excluded. Then the textual features in MD&A section of Chinese listed company's annual reports are extracted using word vector. After that, powerful deep learning models are employed and their performances are compared with numeric data, textual data and combination of them, respectively. The empirical results show great performance improvement of the proposed deep learning methods against traditional machine learning methods, and LSTM, GRU approaches work with testing samples in correct classification rates of 94.98% and 94.62%, indicating that the extracted textual features of MD&A section exhibit promising classification results and substantially reinforce financial fraud detection.

Task II: Write an informative abstract of 200–250 words for the given report.

Report Title: Credit Card Fraud Detection Using State-of-the-Art Machine Learning and Deep Learning Algorithms
(Sourced from F. K. Alarfaj et al.: CCF Detection Using State-of-the-Art ML and DL Algorithms)

Introduction
Credit card fraud is a growing problem worldwide, causing billions of dollars in annual losses. In 2020, over 393,000 fraud cases were reported in the United States alone, making it the second most common type of identity theft. As online transactions increase, so does card-not-present fraud. Financial institutions urgently need automated fraud detection systems that are accurate, fast, and reliable.
However, existing machine learning (ML) methods face several challenges: low accuracy, high false alarm rates, and extreme class imbalance (fraudulent transactions are very rare compared to legitimate ones). Traditional algorithms like Support Vector Machines, Decision Trees, and Logistic Regression have been widely used but do not perform well enough to prevent significant losses.
The objective of this research was to detect credit card fraud more accurately by applying state-of-the-art deep learning algorithms, specifically Convolutional Neural Networks (CNNs), and to compare their performance with traditional machine learning methods.

Methodology

The researchers used a publicly available European credit card dataset from September 2018. It contained 284,807 transactions, of which only 492 (0.172%) were fraudulent. Due to privacy concerns, the dataset features (except Time and Amount) were transformed using Principal Component Analysis (PCA) into 28 variables labeled V1 to V28.

First, six machine learning algorithms were applied: Decision Tree, K-Nearest Neighbours (KNN), Logistic Regression, Support Vector Machine (SVM), Random Forest, and XG Boost. Then, three deep learning architectures based on Convolutional Neural Networks (CNNs) were developed, with layers ranging from 5 to 20. The models were tested using different epoch sizes (14, 35, 50, 100) and both imbalanced and balanced datasets. Performance was measured using accuracy, precision, recall, F1-score, and Area Under the Curve (AUC).

Findings

The machine learning algorithms achieved high accuracy (99.91% to 99.95%) but lower F1-scores (73% to 85%), indicating they struggled with correctly identifying fraudulent transactions. Among ML models, KNN and XG Boost performed best.

The deep learning models significantly improved fraud detection. The proposed CNN with 20 layers, trained over 100 epochs, achieved:

- **Accuracy:** 99.9%
- **F1-score:** 85.71%
- **Precision:** 93%
- **AUC:** 98%

When the dataset was balanced, the CNN model achieved 95.8% validation accuracy. Increasing the number of layers (from 5 to 20) and epochs improved performance. The proposed CNN outperformed both traditional machine learning algorithms and the baseline deep learning model.

Discussion

- **Machine learning algorithms** produced many false negatives (missed frauds) due to class imbalance, even though their overall accuracy appeared high.
- **CNNs with more layers** extracted better features from transaction data, leading to more reliable fraud detection.
- **Balancing the dataset** further improved the model's ability to detect fraudulent transactions without increasing false alarms.
- The **20-layer CNN** was the most effective architecture, balancing precision and recall better than shallower models.

Conclusion and Recommendations

The study concludes that deep learning, particularly Convolutional Neural Networks with multiple layers, is more effective than traditional machine learning for credit card fraud detection. The proposed 20-layer CNN achieved state-of-the-art results with 99.9% accuracy and 98% AUC.

Financial institutions should consider implementing deep learning-based fraud detection systems to reduce financial losses. Future research should explore more advanced deep learning methods, such as generative adversarial networks (GANs) and variational autoencoders (VAEs), and further address the class imbalance problem in real-world transaction data.
